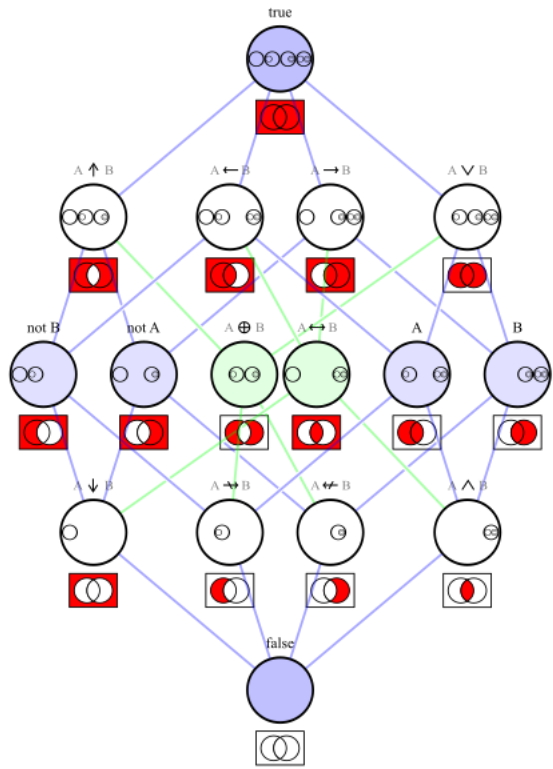


Logical connective

In **logic**, a **logical connective** (also called a **logical operator**, **sentential connective**, or **sentential operator**) is a **logical constant**. Connectives can be used to connect logical formulas. For instance in the **syntax** of **propositional logic**, the **binary** connective $\vee$ can be used to join the two **atomic formulas** *P* and *Q*, rendering the complex formula *P* $\vee$ *Q*.



Hasse diagram of logical connectives.

Common connectives include **negation**, **disjunction**, **conjunction**, **implication**, and **equivalence**. In standard systems of **classical logic**, these connectives are **interpreted** as **truth functions**, though they receive a variety of alternative interpretations in **nonclassical logics**. Their classical interpretations are similar to the meanings of natural language expressions such as **English** "not", "or", "and", and "if", but not identical. Discrepancies between natural language connectives and those of classical logic have motivated nonclassical approaches to natural language meaning as well as approaches which pair a classical **compositional semantics** with a robust **pragmatics**.

A logical connective is similar to, but not equivalent to, a syntax commonly used in programming languages called a **conditional operator**.^[1]

Overview

In **formal languages**, truth functions are represented by unambiguous symbols. This allows logical statements to not be understood in an ambiguous way. These symbols are called *logical connectives*, *logical operators*, *propositional operators*, or, in **classical logic**, *truth-functional connectives*. For the rules which allow new well-formed formulas to be constructed by joining other well-formed formulas using truth-functional connectives, see **well-formed formula**.

Logical connectives can be used to link zero or more statements, so one can speak about ***n**-ary logical connectives*. The **boolean** constants *True* and *False* can be thought of as zero-ary operators. Negation is a 1-ary connective, and so on.

List of common logical connectives

Commonly used logical connectives include the following ones.^[2]

- Negation (not)**: $\neg$, $\sim$, ***N*** (prefix) in which $\neg$ is the most modern and widely used, and $\sim$ is used by many people too;
- Conjunction (and)**: $\wedge$, **&**, ***K*** (prefix) in which $\wedge$ is the most modern and widely used;
- Disjunction (or)**: $\vee$, ***A*** (prefix) in which $\vee$ is the most modern and widely used;
- Implication (if...then)**: $\rightarrow$, $\supset$, $\Rightarrow$, ***C*** (prefix) in which $\rightarrow$ is the most modern and widely used, and $\supset$ is used by many people too;
- Equivalence (if and only if)**: $\leftrightarrow$, $\circ$, $\Leftrightarrow$, $\equiv$, ***E*** (prefix) in which $\leftrightarrow$ is the most modern and widely used, and $\circ$ may be also a good choice compared to $\supset$ denoting implication just like $\leftrightarrow$ to $\rightarrow$.

For example, the meaning of the statements *it is raining* (denoted by ***p***) and *I am indoors* (denoted by ***q***) is transformed, when the two are combined with logical connectives:

- It is ***not*** raining ($\neg \textit{\textbf{p}}$);
- It is raining ***and*** I am indoors ($\textit{\textbf{p}} \wedge \textit{\textbf{q}}$);
- It is raining ***or*** I am indoors ($\textit{\textbf{p}} \vee \textit{\textbf{q}}$);
- If*** it is raining, ***then*** I am indoors ($\textit{\textbf{p}} \rightarrow \textit{\textbf{q}}$);
- If*** I am indoors, ***then*** it is raining ($\textit{\textbf{q}} \rightarrow \textit{\textbf{p}}$);
- I am indoors ***if and only if*** it is raining ($\textit{\textbf{p}} \leftrightarrow \textit{\textbf{q}}$).

It is also common to consider the *always true* formula and the *always false* formula to be connective (in which case they are **nullary**).

- True** formula: $\top$, **1**, ***V*** (prefix), or **T**;
- False** formula: $\perp$, **0**, ***O*** (prefix), or **F**.

This table summarizes the terminology:

Connective	In English	Noun for parts	Verb phrase
Conjunction	Both A and B	conjunct	A and B are conjoined
Disjunction	Either A or B, or both	disjunct	A and B are disjoined
Negation	It is not the case that A	negatum/negand	A is negated
Conditional	If A, then B	antecedent, consequent	B is implied by A
Biconditional	A if, and only if, B	equivalents	A and B are equivalent

History of notations

- Negation: the symbol $\neg$ appeared in **Heyting** in 1930^{[3][4]} (compare to **Frege**'s symbol $\neg$ in his **Begriffsschrift**^[5]); the symbol $\sim$ appeared in **Russell** in 1908;^[6] an alternative notation is to add a horizontal line on top of the formula, as in $\overline{\textit{\textbf{p}}}$; another alternative notation is to use a **prime symbol** as in $\textit{\textbf{p}}'$.
- Conjunction: the symbol $\wedge$ appeared in Heyting in 1930^[3] (compare to **Peano**'s use of the set-theoretic notation of **intersection** $\cap$ ^[7]); the symbol **&** appeared at least in **Schönfinkel** in 1924;^[8] the symbol ***** comes from **Boole**'s interpretation of logic as an **elementary algebra**.

- Disjunction: the symbol ∨ appeared in [Russell](#) in 1908^[6] (compare to [Peano](#)'s use of the set-theoretic notation of [union](#) ∪); the symbol + is also used, in spite of the ambiguity coming from the fact that the + of ordinary [elementary algebra](#) is an [exclusive or](#) when interpreted logically in a two-element [ring](#); punctually in the history a + together with a dot in the lower right corner has been used by [Peirce](#).^[9]
- Implication: the symbol → appeared in [Hilbert](#) in 1918;^{[10]:76} ⊃ was used by Russell in 1908^[6] (compare to Peano's ⊃ the inverted C); ⇒ appeared in [Bourbaki](#) in 1954.^[11]
- Equivalence: the symbol ≡ in [Frege](#) in 1879;^[12] ↔ in Becker in 1933 (not the first time and for this see the following);^[13] ⇔ appeared in [Bourbaki](#) in 1954;^[14] other symbols appeared punctually in the history, such as ⊃⊂ in [Gentzen](#),^[15] ∼ in Schönfinkel^[8] or ⊂⊃ in Chazal,^[16]
- True: the symbol **1** comes from [Boole](#)'s interpretation of logic as an [elementary algebra](#) over the [two-element Boolean algebra](#); other notations include **⋁** (abbreviation for the Latin word "verum") to be found in Peano in 1889.
- False: the symbol **0** comes also from Boole's interpretation of logic as a ring; other notations include **⋀** (rotated **⋁**) to be found in Peano in 1889.

Some authors used letters for connectives: **u.** for conjunction (German's "und" for "and") and **o.** for disjunction (German's "oder" for "or") in early works by Hilbert (1904);^[17] ***Np*** for negation, ***Kpq*** for conjunction, ***Dpq*** for alternative denial, ***Apq*** for disjunction, ***Cpq*** for implication, ***Epq*** for biconditional in [Łukasiewicz](#) in 1929.

Redundancy

Such a logical connective as [converse implication](#) "←" is actually the same as [material conditional](#) with swapped arguments; thus, the symbol for converse implication is redundant. In some logical calculi (notably, in [classical logic](#)), certain essentially different compound statements are [logically equivalent](#). A less [trivial](#) example of a redundancy is the classical equivalence between ¬*p* ∨ *q* and *p* → *q*. Therefore, a classical-based logical system does not need the conditional operator "→" if "¬" (not) and "∨" (or) are already in use, or may use the "→" only as a [syntactic sugar](#) for a compound having one negation and one disjunction.

There are sixteen [Boolean functions](#) associating the input [truth values](#) *p* and *q* with four-digit [binary](#) outputs.^[18] These correspond to possible choices of binary logical connectives for [classical logic](#). Different implementations of classical logic can choose different [functionally complete](#) subsets of connectives.

One approach is to choose a *minimal* set, and define other connectives by some logical form, as in the example with the material conditional above. The following are the [minimal functionally complete sets of operators](#) in classical logic whose arities do not exceed 2:

- One element**
{↑}. {↓}.
- Two elements**
{∨, ¬}, {∧, ¬}, {→, ¬}, {←, ¬}, {→, ⊥}, {←, ⊥}, {→, ↔}, {←, ↔}, {→, ↗}, {→, ↔}, {←, ↗}, {←, ↔}, {↗, ¬}, {↗, ¬}, {↗, ⊤}, {↗, ⊤}, {↗, ↔}, {↗, ↔}.
- Three elements**
{∨, ↔, ⊥}, {∨, ↔, ↔}, {∨, ↔, ⊤}, {∧, ↔, ⊥}, {∧, ↔, ↔}, {∧, ↔, ⊤}.

Another approach is to use with equal rights connectives of a certain convenient and functionally complete, but *not minimal* set. This approach requires more propositional [axioms](#), and each equivalence between logical forms must be either an [axiom](#) or provable as a theorem.

The situation, however, is more complicated in [intuitionistic logic](#). Of its five connectives, {∧, ∨, →, ¬, ⊥}, only negation "¬" can be reduced to other connectives (see [False \(logic\) § False, negation and contradiction](#) for more). Neither conjunction, disjunction, nor material conditional has an equivalent form constructed from the other four logical connectives.

Natural language

The standard logical connectives of classical logic have rough equivalents in the grammars of natural languages. In [English](#), as in many languages, such expressions are typically [grammatical conjunctions](#). However, they can also take the form of [complementizers](#), [verb suffixes](#), and [particles](#). The [denotations](#) of natural language connectives is a major topic of research in [formal semantics](#), a field that studies the logical structure of natural languages.

The meanings of natural language connectives are not precisely identical to their nearest equivalents in classical logic. In particular, disjunction can receive an [exclusive interpretation](#) in many languages. Some researchers have taken this fact as evidence that natural language [semantics](#) is [nonclassical](#). However, others maintain classical semantics by positing [pragmatic](#) accounts of exclusivity which create the illusion of nonclassicality. In such accounts, exclusivity is typically treated as a [scalar implicature](#). Related puzzles involving disjunction include [free choice inferences](#), [Hurford's Constraint](#), and the contribution of disjunction in [alternative questions](#).

Other apparent discrepancies between natural language and classical logic include the [paradoxes of material implication](#), [donkey anaphora](#) and the problem of [counterfactual conditionals](#). These phenomena have been taken as motivation for identifying the denotations of natural language conditionals with logical operators including the [strict conditional](#), the [variably strict conditional](#), as well as various [dynamic](#) operators.

The following table shows the standard classically definable approximations for the English connectives.

English word	Connective	Symbol	Logical gate
not	negation	¬	NOT
and	conjunction	∧	AND
or	disjunction	∨	OR
if...then	material implication	→	IMPLY
...if	converse implication	←	
either...or	exclusive disjunction	⊕	XOR
if and only if	biconditional	↔	XNOR
not both	alternative denial	↑	NAND
neither...nor	joint denial	↓	NOR
but not	material nonimplication	↗	NIMPLY

Properties

Some logical connectives possess properties that may be expressed in the theorems containing the connective. Some of those properties that a logical connective may have are:

Associativity

Within an expression containing two or more of the same associative connectives in a row, the order of the operations does not matter as long as the sequence of the operands is not changed.

Commutativity

The operands of the connective may be swapped, preserving logical equivalence to the original expression.

Distributivity

Symbol, name	Truth table	Venn diagram		
Zeroary connectives (constants)				
⊤ Truth/tautology	1			
⊥ Falsity/contradiction	0			
Unary connectives				
$p =$	0	1		
Proposition p	0	1 		
¬ Negation	1	0 		
Binary connectives				
$p =$	0	1		
$q =$	0	1		
Proposition p	0	0	1	1 
Proposition q	0	1	0	1 
∧ Conjunction	0	0	0	1 
↑ Alternative denial	1	1	1	0 
∨ Disjunction	0	1	1	1 
↓ Joint denial	1	0	0	0 
→ Material conditional	1	1	0	1 
↔ Exclusive or	0	1	1	0 
↔ Biconditional	1	0	0	1 
← Converse implication	1	0	1	1 
More information				

A connective denoted by $\cdot$ distributes over another connective denoted by $+$, if $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$ for all operands a, b, c .

Idempotence

Whenever the operands of the operation are the same, the compound is logically equivalent to the operand.

Absorption

A pair of connectives $\wedge, \vee$ satisfies the absorption law if $a \wedge (a \vee b) = a$ for all operands a, b .

Monotonicity

If $f(a_1, ..., a_n) \leq f(b_1, ..., b_n)$ for all $a_1, ..., a_n, b_1, ..., b_n \in \{0,1\}$ such that $a_1 \leq b_1, a_2 \leq b_2, ..., a_n \leq b_n$. E.g., $\vee, \wedge, \top, \perp$.

Affinity

Each variable always makes a difference in the truth-value of the operation or it never makes a difference. E.g., $\neg, \leftrightarrow, \leftrightarrow, \top, \perp$.

Duality

To read the truth-value assignments for the operation from top to bottom on its **truth table** is the same as taking the complement of reading the table of the same or another connective from bottom to top. Without resorting to truth tables it may be formulated as $\tilde{g}(\neg a_1, ..., \neg a_n) = \neg g(a_1, ..., a_n)$. E.g., $\neg$.

Truth-preserving

The compound all those arguments are tautologies is a tautology itself. E.g., $\vee, \wedge, \top, \rightarrow, \leftrightarrow, \subset$ (see **validity**).

Falsehood-preserving

The compound all those argument are **contradictions** is a contradiction itself. E.g., $\vee, \wedge, \leftrightarrow, \perp, \not\subset, \supset$ (see **validity**).

Involutivity (for unary connectives)

$f(f(a)) = a$. E.g. negation in classical logic.

For classical and intuitionistic logic, the "=" symbol means that corresponding implications "... $\rightarrow$..." and "... $\leftarrow$..." for logical compounds can be both proved as theorems, and the " $\leq$ " symbol means that "... $\rightarrow$..." for logical compounds is a consequence of corresponding "... $\rightarrow$..." connectives for propositional variables. Some **many-valued logics** may have incompatible definitions of equivalence and order (entailment).

Both conjunction and disjunction are associative, commutative and idempotent in classical logic, most varieties of many-valued logic and intuitionistic logic. The same is true about distributivity of conjunction over disjunction and disjunction over conjunction, as well as for the absorption law.

In classical logic and some varieties of many-valued logic, conjunction and disjunction are dual, and negation is self-dual, the latter is also self-dual in intuitionistic logic.

Order of precedence

As a way of reducing the number of necessary parentheses, one may introduce **precedence rules**: $\neg$ has higher precedence than $\wedge$, $\wedge$ higher than $\vee$, and $\vee$ higher than $\rightarrow$. So for example, $P \vee Q \wedge \neg R \rightarrow S$ is short for $(P \vee (Q \wedge (\neg R))) \rightarrow S$.

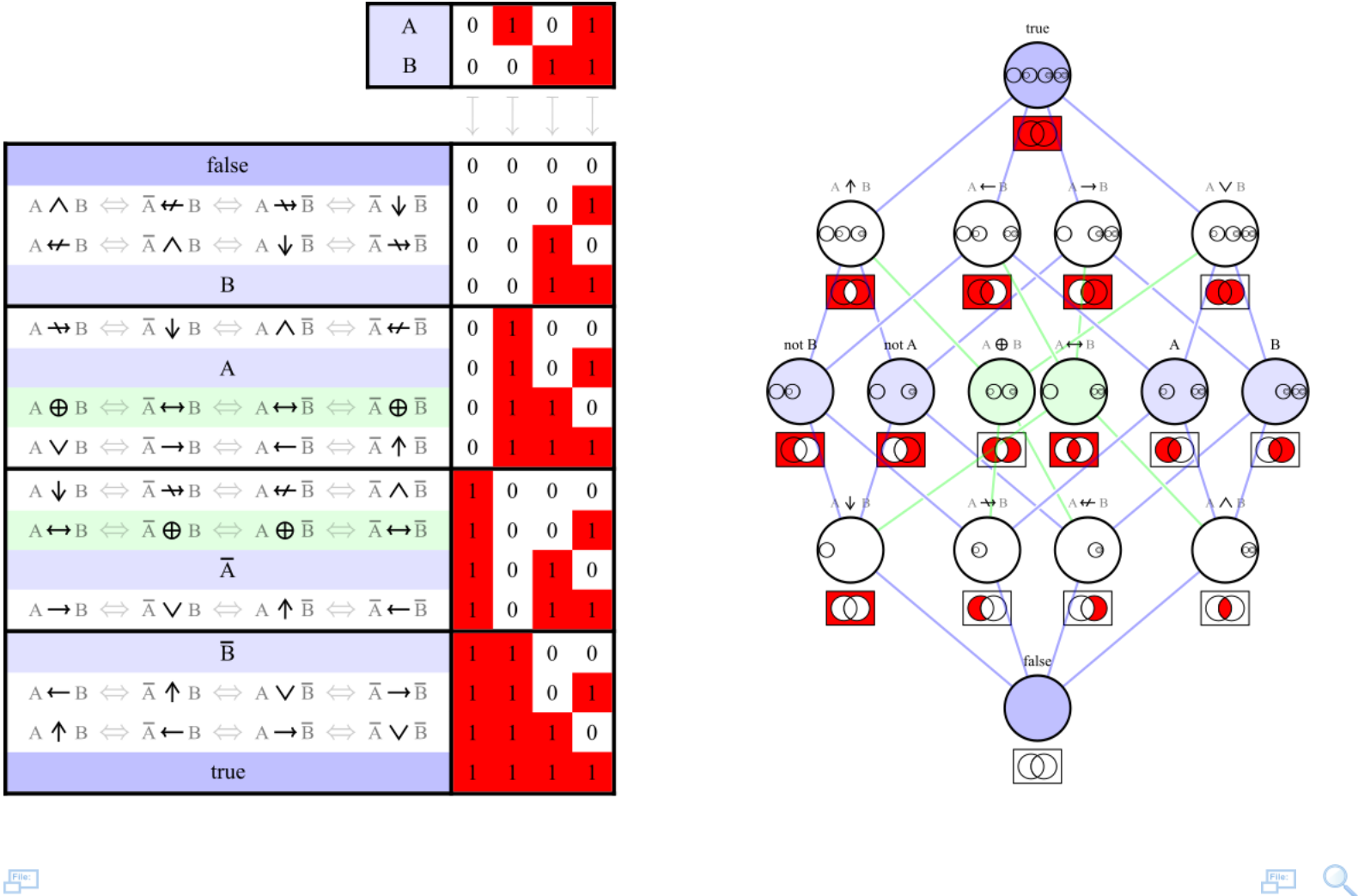
Here is a table that shows a commonly used precedence of logical operators.^{[19][20]}

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

However, not all compilers use the same order; for instance, an ordering in which disjunction is lower precedence than implication or bi-implication has also been used.^[21] Sometimes precedence between conjunction and disjunction is unspecified requiring to provide it explicitly in given formula with parentheses. The order of precedence determines which connective is the "main connective" when interpreting a non-atomic formula.

Table and Hasse diagram

The 16 logical connectives can be **partially ordered** to produce the following **Hasse diagram**. The partial order is defined by declaring that $x \leq y$ if and only if whenever x holds then so does y .



Applications

Logical connectives are used in **computer science** and in **set theory**.

Computer science

A truth-functional approach to logical operators is implemented as **logic gates** in **digital circuits**. Practically all digital circuits (the major exception is **DRAM**) are built up from **NAND**, **NOR**, **NOT**, and **transmission gates**; see more details in **Truth function in computer science**. Logical operators over **bit vectors** (corresponding to finite **Boolean algebras**) are **bitwise operations**.

- [Rautenberg, W.](#) (2010), *A Concise Introduction to Mathematical Logic* (3rd ed.), New York: Springer Science+Business Media, doi:10.1007/978-1-4419-1221-3 (<https://doi.org/10.1007%2F978-1-4419-1221-3>) , ISBN 978-1-4419-1220-6.
- Humberstone, Lloyd (2011). *The Connectives*. MIT Press. ISBN 978-0-262-01654-4.

External links

- ["Propositional connective"](https://www.encyclopediaofmath.org/index.php?title=Propositional_connective) (https://www.encyclopediaofmath.org/index.php?title=Propositional_connective) , *Encyclopedia of Mathematics*, EMS Press, 2001 [1994]
- Lloyd Humberstone (2010), "[Sentence Connectives in Formal Logic](https://plato.stanford.edu/entries/connectives-logic/) (<https://plato.stanford.edu/entries/connectives-logic/>) " , *Stanford Encyclopedia of Philosophy* (An [abstract algebraic logic](#) approach to connectives.)
- John MacFarlane (2005), "[Logical constants](https://plato.stanford.edu/entries/logical-constants/) (<https://plato.stanford.edu/entries/logical-constants/>) " , *Stanford Encyclopedia of Philosophy*.